

Rajendran Prathibha Devkar

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EXPERIENCE

Software Developer Intern

May 2025 - Mar 2026

Board Americas - Remote, USA

- Delivered 40+ production fixes across an enterprise forecasting platform (Aurelia/TypeScript, C#), improving release cadence and reducing hot fixes after release.
- Diagnosed and resolved critical D3.js data visualization defects including fiscal/calendar year logic errors and actual versus forecast rendering issues affecting downstream decision-making.
- Overhauled Workbench modules and fixed a cache invalidation bug, eliminating stale session states and improving real time data sync for concurrent users.
- Implemented validation and metadata fixes across indicator workflows, preventing invalid configuration states and blocking incorrect downstream actions in forecasting pipelines.

Technology Analyst

Oct 2019 - Nov 2022

Infosys Private Ltd. - Trivandrum, India

- Improved user satisfaction by 40% by architecting a React Native curbside pickup app that reduce checkout friction and scheduling steps for store fulfillment.
- Designed and optimized data schemas and query pipelines supporting 1,000+ concurrent users, cutting retrieval latency by 40% and enabling reliable session analytics at scale.
- Boosted application performance by 20% by designing and executing JMeter load tests to simulate peak traffic, identifying and resolving architectural bottlenecks prior to deployment.
- Resolved a critical multi-user production crash caused by a Calendar API race condition under concurrent load, restoring platform stability and ensuring 98% uptime via PM2 multi-server rollouts.

SKILLS

ML / DATA: PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas

AI / LLM : RAG, LangChain, PineCone, Prompt Engineering

Languages: Python, TypeScript, JavaScript, C#, Java, C, C++

Backend / APIs: ASP.Net, Node.js, FastAPI, RESTful APIs

Cloud: AWS, GCP

IoT / Streaming: MQTT, InfluxDB, Grafana

Databases: SQL Server, PostgreSQL, MySQL

Frontend: React, React Native, Aurelia, D3.js, HTML5, CSS3

DevOps / Tools: Azure DevOps, Git, Docker, Jira

Testing: Regression, API, Performance (JMeter)

PROJECTS

Recurrent Depth Transformer (RDT) / PyTorch, Python

- Implemented a looped weight sharing transformer achieving equivalent reasoning depth to a standard GPT at 47% fewer parameters, trained entirely from scratch on consumer hardware.
- Fine-tuned the model into a tool calling agent using chain of thought reasoning, benchmarked across 3 tasks, showing looped inference can match stacked depth without retraining.

Intelligent Handwritten Document Recognition (HTR) / YOLOv8, TrOCR, Python

- Trained a YOLOv8s layout detector (76.11% mAP@50) and integrated TrOCR-large (33% CER) to build an end to end pipeline exporting structured Markdown and JSON from handwritten documents.

COVID-19 Detection via Chest X-Rays / EfficientNet-B3, ResNet-50, PyTorch, Streamlit

- Achieved 93.16% test accuracy using EfficientNet-B3 on the COVID-QU-Ex dataset (~33.9K images); deployed as a real time Streamlit inference app.

HVAC Lab Energy Monitoring Platform / Python, MQTT, InfluxDB, Grafana, Docker

- Built a real time IoT streaming layer for HVAC, water heater, and solar telemetry using MQTT pub/sub, with normalized pipelines (Cel, Watts) for InfluxDB storage and Grafana visualization.

EDUCATION

Master of Science in Computer Science - The University of Texas at Arlington, USA

May 2026

Bachelor of Technology in Computer Science - SaintGits College of Engineering, India

May 2019

PUBLICATION

Devkar, R. Prathibha, Daniel, M.S, John, N.P, et al. (2021). Speaking Mouth System for Speech-Impaired People using Hand Gestures

- Developed a CNN-based real time gesture recognition system (96% accuracy) converting hand gestures into text and synthesized speech; received KSCSTE INNOVATE research funding.

CERTIFICATION

Hewlett Packard Enterprise Software Engineering - Forage

Apr 2025

IBM Data Science Professional Certificate - Coursera

May 2024